

# Tide retrospective: *cumulants as derivatives of the log-partition*

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## Abstract

The synthesis that closes the anharmonic cumulant arc: the connected cumulants  $\kappa_2, \kappa_3, \kappa_4$  — proved across the arc as derivatives of the perturbed *mean*  $M = G_1/G_0$  — are identified as derivatives of the genuine cumulant generating function  $K = \log Z$  (with  $Z = G_0$ ):

$$\begin{aligned} \text{iteratedDeriv } (n+1) K 0 &= \text{iteratedDeriv } n M 0, \\ \text{so } \kappa_2 &= \partial^2 K|_0, \quad \kappa_3 = \partial^3 K|_0, \quad \kappa_4 = \partial^4 K|_0. \end{aligned}$$

The new file<sup>1</sup> builds clean first try, no `sorry`/`axiom`/`native_decide`.

## 1 Setting

The tenth and final tide of a single-day anharmonic arc on the laplace seabed. Across the arc the cumulants were computed as derivatives of the mean  $M(h) = \langle u \rangle_h = G_1(h)/G_0(h)$  (because  $K' = M$ , the fluctuation–dissipation route is the most direct). This tide records the “why”:  $K = \log Z$  is the cumulant *generating* function, so its derivatives at 0 *are* the connected cumulants. Closing this loop — by relating  $\partial^{n+1} \log Z$  to  $\partial^n M$  — restores the textbook definition and ties the whole arc back to its first rung, `logPartition_hasDerivAt`.

## 2 The thing formalised

Writing  $K(h) = \log Z(h) = \log G_0(h)$  and  $M(h) = G_1(h)/G_0(h)$ :

*For the anharmonic potential,  $\lambda, \gamma > 0$ ,  $\alpha^2 < 3\lambda\gamma$ ,  $t > 0$ :  $\text{iteratedDeriv } (n+1) K 0 = \text{iteratedDeriv } n M 0$  for all  $n$ ; and the named corollaries  $\partial^2 K|_0 = \kappa_2$ ,  $\partial^3 K|_0 = \kappa_3$ ,  $\partial^4 K|_0 = \kappa_4$  (each in the explicit  $G_0, \dots, G_4$  closed form of the corresponding cumulant tide).*

## 3 Proof strategy

Three short steps. (i) The general- $h$  first derivative of  $K$  is the mean: `HasDerivAt.log` on the partition derivative `HasDerivAt  $G_0(G_1(h_0)) h_0$`  with  $G_0(h_0) > 0$  gives  $K'(h_0) = G_1(h_0)/G_0(h_0)$ , hence  $\text{deriv } K =_{\text{ev}} M$  on `ball 0 1`. (ii) The ladder lemma: by `iteratedDeriv_succ`,  $\text{iteratedDeriv } (n+1) K = \text{iteratedDeriv } n (\text{deriv } K)$ , and `Filter.EventuallyEq.iteratedDeriv_eq`

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<sup>1</sup>`AnharmonicCumulantsLogPartition.lean`.

rewrites `deriv K` to `M`, giving `iteratedDeriv (n+1) K 0 = iteratedDeriv n M 0`. (iii) The three corollaries are `rw [ladder]` then `exact` the existing cumulant theorems ( $\kappa_2$  at  $n = 1$  via `iteratedDeriv_one`,  $\kappa_3$  at  $n = 2$ ,  $\kappa_4$  at  $n = 3$ ).

## 4 Roadblocks and resolutions

None — first-try build. The synthesis is pure chaining of established lemmas, and crucially `Filter.EventuallyEq.iteratedDeriv_eq` already exists in `Mathlib`, so the “derivatives respect germs” step needed no bespoke induction. Because the corollaries discharge by `exact`-ing the cumulant theorems, Lean checks that the stated closed forms match the proved ones — no risk of a silently mistyped  $\kappa_4$ .

## 5 What was Mathlib, what was new

**Mathlib.** `HasDerivAt.log`, `HasDerivAt.deriv`; `iteratedDeriv_succ'`, `iteratedDeriv_one`; `Filter.EventuallyEq.iteratedDeriv_eq`.

**Seabed (reused).** The partition derivative and positivity, and the three cumulant theorems — all from this session’s arc.

**New.** The general- $h$  log-partition derivative, the ladder lemma, and the three  $\partial^n \log Z$  corollaries. About 130 lines.

## 6 Lessons

- **Two routes to the cumulants, one cheap bridge.** Computing  $\kappa_n$  as  $\partial^{n-1} M$  (fluctuation-dissipation) sidesteps the log entirely and was the right call for the heavy rungs; a single ladder lemma then relabels them as  $\partial^n \log Z$  for the textbook statement.
- **`EventuallyEq.iteratedDeriv_eq` is the clean tool for “germ-local” iterated derivatives.** It turns one eventual equality ( $\text{deriv } K =_{\text{ev}} M$ ) into equality of all iterated derivatives at the point, no induction required.

## 7 Follow-ups

- **Moment-form corollaries.** Express each  $\kappa_n$  in the  $\langle u^k \rangle_0$  (`gibbsExp`) form via moment-normalisation ( $\kappa_2$  already done in the second-cumulant tide).
- **Higher cumulants**  $\kappa_5, \dots$  if wanted: each is one more identical differentiation layer.
- **Promote** `amgm_t_abs_x` to `lean-common`.